

FRONTLINE HCP MODULE

Prostate Cancer

Your interactive learning journey: recognise, discuss, refer, understand, support.

English

Rwanda Biomedical Centre | Frontline HCP

Module objectives



Recognise

Recognise what is prostate cancer and its characteristics.



Discuss

Discuss early detection in a balanced way and understand PSA.



Refer

Refer correctly and know the urgent warning signs.



Understand diagnosis

Understand how a diagnosis is confirmed, and what stage and grade mean.



Understand treatment

Understand treatment options and their common effects.



Support

Support follow-up, quality of life, and palliative care.

What is inside

Pre-test

Twenty short questions to check your starting point. Not graded for you.

Session 1: Understanding

What prostate cancer is, how common it is, and what stage and grade mean.

Session 2: Screening

Who is at risk, what to look for, and the balance of PSA-based early detection.

Session 3: Referral and counselling

How urgently to refer, red flags, and how to counsel with care.

Session 4: Diagnosis and treatment

How a diagnosis is confirmed and an overview of treatment.

Session 5: Supportive care

Effects to recognise, emergencies to escalate, and quality of life.

References

The sources used, including GLOBOCAN 2024.

Pre-test

BASELINE CHECK

Knowledge check (1/10)

Q1. Most prostate cancers start in which cells?

- A. Muscle cells
- B. Glandular (epithelial) cells
- C. Nerve cells
- D. Blood vessels

Q2. Which is an established risk factor?

- A. Younger age
- B. Family history of prostate cancer
- C. High water intake
- D. Living at altitude

BASELINE CHECK

Knowledge check (2/10)

Q3. Early prostate cancer usually causes:

- A. Severe pain
- B. Often no symptoms
- C. Immediate weight loss
- D. Fever

Q4. A raised prostate specific antigen (PSA) blood test means:

- A. Cancer is confirmed
- B. Further assessment may be needed
- C. No further action
- D. A benign result only

BASELINE CHECK

Knowledge check (3/10)

Q5. Urinary symptoms such as a weak stream:

- A. Confirm prostate cancer
- B. Can have benign causes
- C. Are always harmless
- D. Always mean infection

Q6. Prostate cancer commonly spreads to:

- A. Bone
- B. Skin
- C. Hair
- D. Teeth

BASELINE CHECK

Knowledge check (4/10)

Q7. Stage describes:

- A. How aggressive the cells look
- B. Where the cancer is and how far it has spread
- C. The PSA value
- D. The treatment chosen

Q8. Grade (Gleason or Grade Group) describes:

- A. Extent of spread
- B. How abnormal the cells look
- C. The patient's age
- D. The PSA level

BASELINE CHECK

Knowledge check (5/10)

Q9. A normal digital rectal examination:

- A. Excludes cancer
- B. Does not exclude cancer
- C. Confirms cancer
- D. Replaces biopsy

Q10. Prostate cancer is confirmed by:

- A. PSA alone
- B. Symptoms alone
- C. Biopsy and histopathology
- D. Examination alone

BASELINE CHECK

Knowledge check (6/10)

Q11. Screening applies to:

- A. People with symptoms
- B. Asymptomatic people considering early detection
- C. Only people with cancer
- D. No one

Q12. Active surveillance means:

- A. Ignoring the cancer
- B. Structured monitoring of suitable low-risk cancer
- C. Immediate surgery
- D. Palliative care only

BASELINE CHECK

Knowledge check (7/10)

Q13. A red flag needing urgent action is:

- A. Mild nocturia
- B. New leg weakness or loss of bladder or bowel control
- C. Occasional frequency
- D. Mild hesitancy

Q14. Androgen deprivation therapy works by:

- A. Adding testosterone
- B. Reducing or blocking androgen signalling
- C. Killing all cells equally
- D. Only relieving pain

BASELINE CHECK

Knowledge check (8/10)

Q15. Overdiagnosis means:

- A.** Missing a cancer
- B.** Detecting a cancer that would not have caused harm
- C.** A wrong diagnosis
- D.** A late diagnosis

Q16. Radiotherapy:

- A.** Uses ionizing radiation to damage cancer cell DNA
- B.** Is only a type of surgery
- C.** Adds hormones
- D.** Has no side effects

BASELINE CHECK

Knowledge check (9/10)

Q17. Which is often used to assess possible bone spread?

- A. Ultrasound of the neck
- B. Bone imaging
- C. Skin scan
- D. No imaging

Q18. Counselling before referral should:

- A. Promise a diagnosis
- B. Explain that assessment is needed without creating fear
- C. Withhold information
- D. Avoid the patient's questions

BASELINE CHECK

Knowledge check (10/10)

Q19. Palliative care is:

- A. Only for the last days of life
- B. Symptom relief and quality of life, alongside treatment when appropriate
- C. The same as stopping care
- D. Only pain medicine

Q20. In Rwanda, prostate cancer in men is:

- A. Rare
- B. The most common cancer in men in current estimates
- C. Only found abroad
- D. Not recorded

Session 1

Understanding prostate cancer

Build the foundation: what prostate cancer is, how common it is, and what stage and grade mean.

SESSION 1

Learning objectives



Describe

Describe what prostate cancer is and where it develops.



Interpret

Interpret the global, Africa and Rwanda burden, including projections.



Explain anatomy

Explain prostate anatomy relevant to symptoms and examination.



Outline

Outline how prostate cancer develops and spreads to bone.



Distinguish

Explain TNM and distinguish stage from grade.



Describe grade

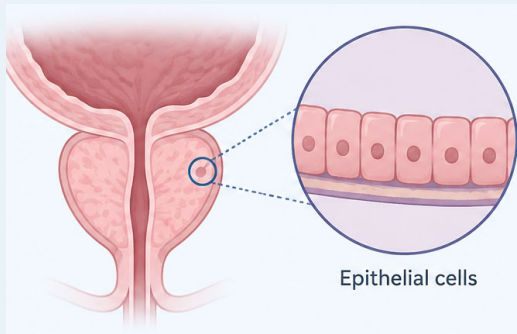
Describe Gleason score and Grade Groups at a high level.

SESSION 1

What prostate cancer is

Cellular origin

Most prostate cancers are adenocarcinomas, arising from the gland's epithelial cells.



Behaviour varies

Some cancers grow slowly and stay silent; others are more aggressive.



Slow-growing
(indolent)



May stay silent
for years



Fast-growing
(aggressive)



More likely to
spread and cause
problems

Why early detection

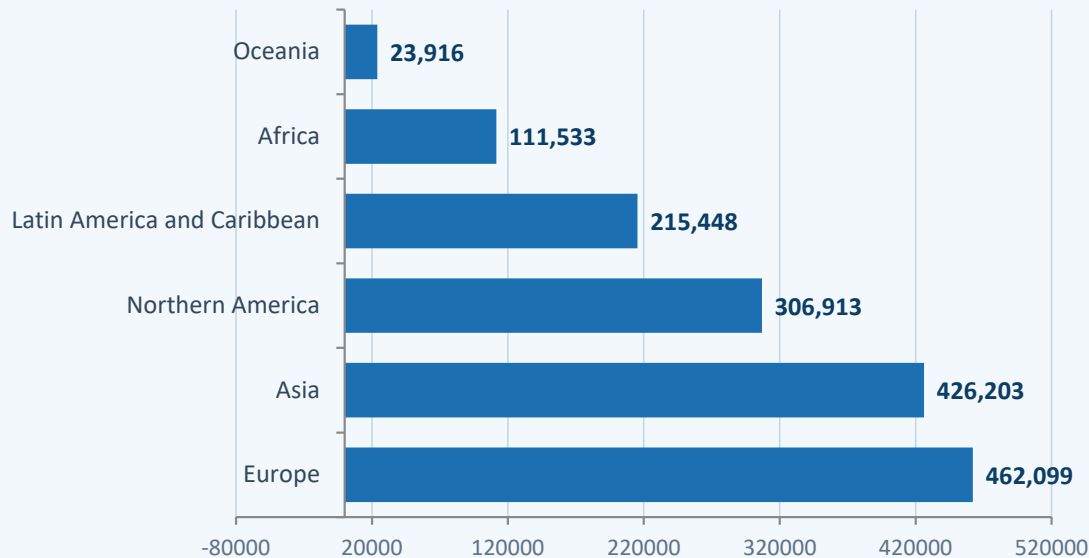
Finding clinically important cancer earlier can widen the options for care.

An abnormal screening or examination result does not on its own confirm cancer.

SESSION 1

How common it is worldwide

New prostate cancer cases by world region, 2024



Source: GLOBOCAN 2024 (IARC), Cancer Today, 2024 estimates version 1.0.

1,546,112

New cases in men worldwide, 2024

419,849

Deaths in men worldwide, 2024

2nd

Most common cancer in men worldwide

SESSION 1

Africa and Rwanda

111,533

New cases in Africa, 2024 (7.2% of the global total)

61,796

Deaths in Africa, 2024 (14.7% of the global total)

~174%

Projected rise by 2050, to about 305,000

1st

Most common cancer in men in Rwanda, 2024 estimates

1,856

Estimated new cases in Rwanda, 2024

1,179

Estimated deaths in Rwanda, 2024

SESSION 1

Prostate anatomy

Location

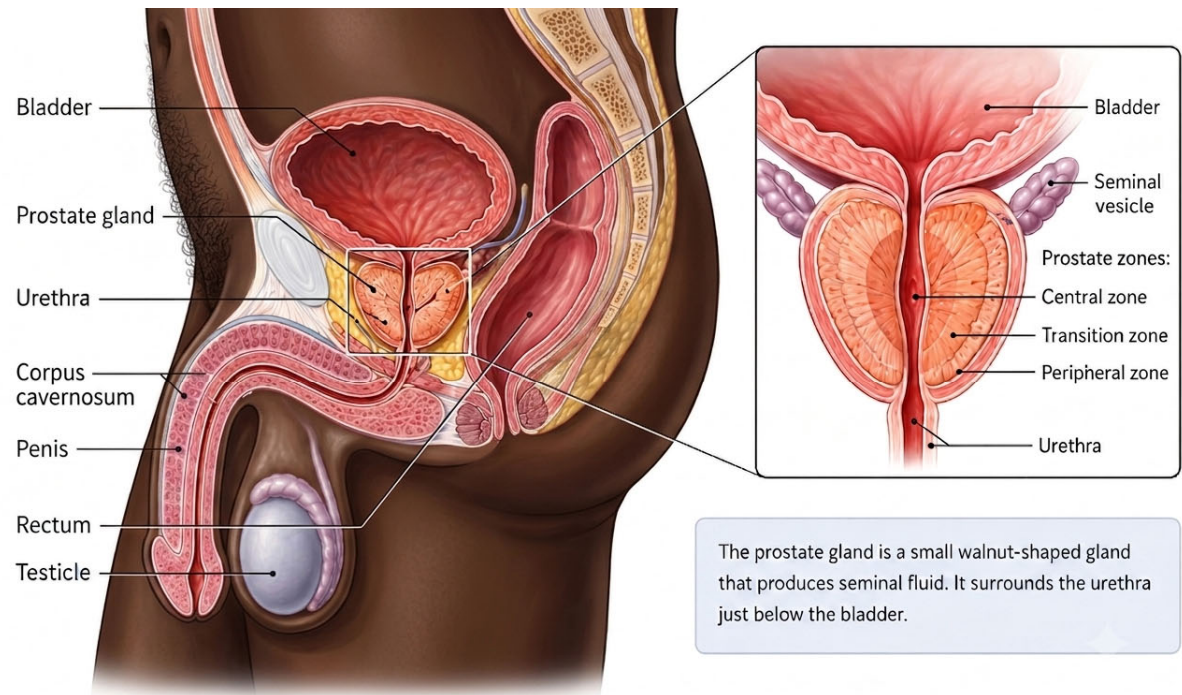
A small gland below the bladder, in front of the rectum, around the first part of the urethra.

Urethra

The urethra runs through the prostate, so disease can affect the urinary stream.

Rectum and examination

The back of the gland lies next to the rectum, so a digital rectal examination can assess it.



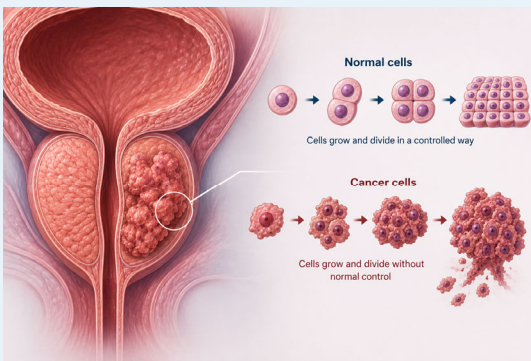
Local growth: Tumour growth may extend beyond the gland to nearby structures over time.

SESSION 1

How it develops

Abnormal growth

Cancer begins when prostate cells grow and divide without normal control.



Type

Most prostate cancers are adenocarcinomas.



Local, then beyond

Disease may stay local, then extend beyond the gland. It can spread by lymph or blood, commonly to bone.



SESSION 1

TNM

T: primary tumour

Describes whether disease is confined to the prostate, extends beyond it, or invades nearby structures.

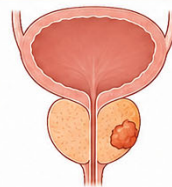
N: lymph nodes

Describes whether regional lymph nodes are involved.

M: distant spread

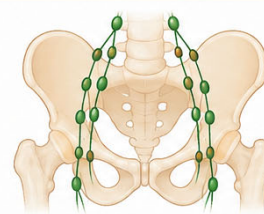
Describes distant spread. Prostate cancer commonly spreads to bone.

T – TUMOR (Primary)



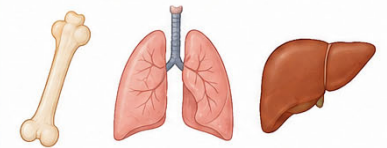
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
T1	Tumor not visible or not felt
T2	Tumor is in the prostate
T3	Tumor has grown outside the prostate
T4	Tumor has spread to nearby structures (other than seminal vesicles)

N – NODES (Regional)



NX	Regional lymph nodes cannot be assessed
N0	No spread to regional lymph nodes
N1	Spread to regional lymph nodes

M – METASTASIS (Distant)

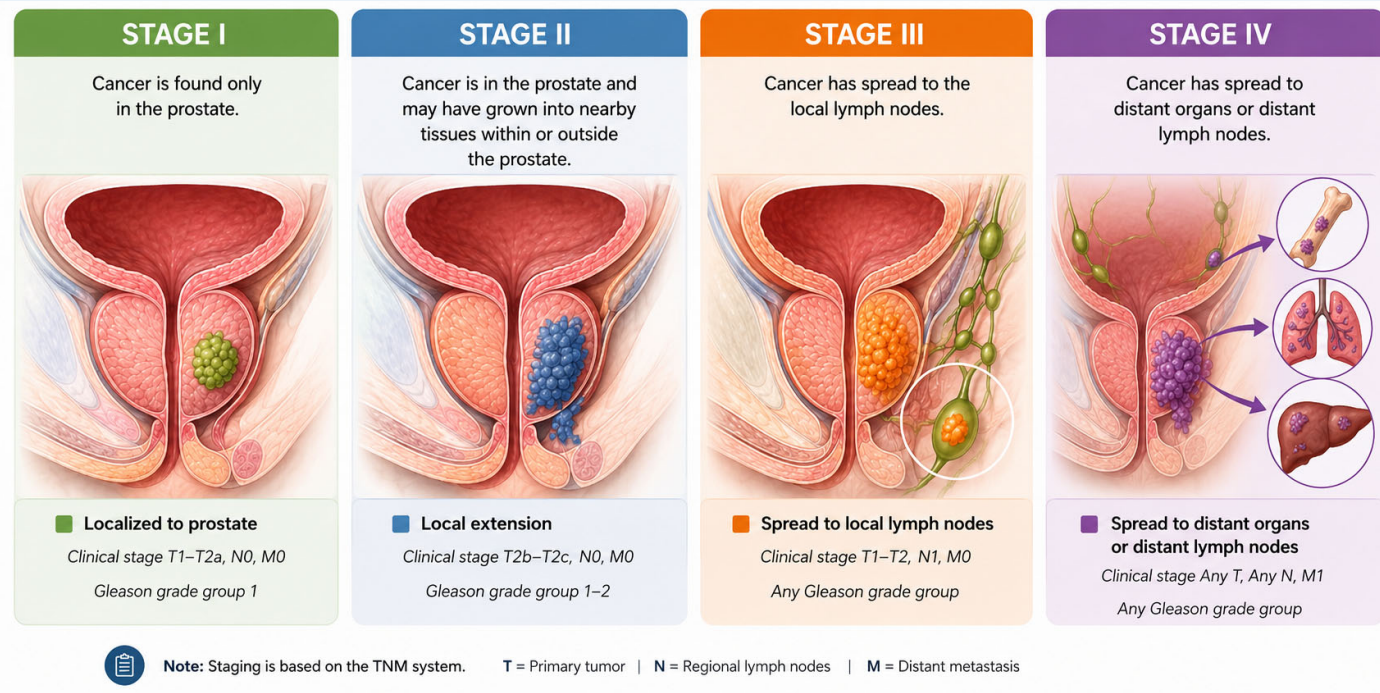


M0	No distant metastasis
M1	Distant metastasis present

SESSION 1

Stage

Quantifies the anatomical extent of a malignant disease using standardized criteria to guide therapeutic decision-making and prognosis assessment.



SESSION 1

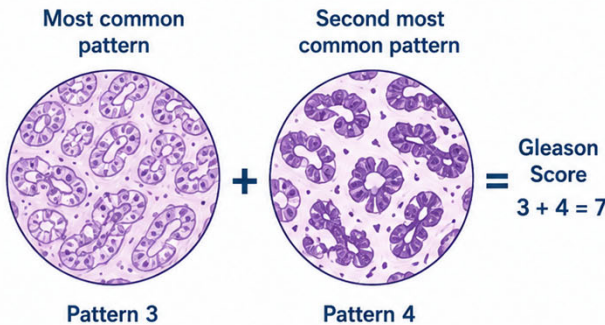
Grade

Gleason and Grade Groups

A pathologist grades the tissue. Lower Grade Groups are less aggressive; higher are more aggressive. You do not grade it yourself

GLEASON SCORE

Looks at how abnormal the cancer cells look under the microscope.



GRADE GROUPS

Groups the Gleason score to show how aggressive the cancer is.

Grade Group	Gleason Score	What it means
1	6 (3+3)	Least aggressive
2	7 (3+4)	Low risk
3	7 (4+3)	Intermediate risk
4	8 (4+4, 3+5, 5+3)	High risk
5	9-10 (4+5, 5+4, 5+5)	Highest risk

SESSION 1

Check your understanding

Which sequence best reflects how prostate cancer typically progresses?

- A** Distant spread, then local disease, then abnormal growth
- B** Abnormal growth, local disease, local extension, then distant spread ✓
- C** Local extension first, with no earlier confined phase
- D** Symptoms always appear before any abnormal growth

Correct answer: B

Why Disease usually begins as abnormal growth within the gland, may extend locally, and can later spread, commonly to bone. Early disease is often silent.

SESSION 1

Knowledge check (1/3)

Q1. Most prostate cancers arise from which cell type?

A. Glandular (epithelial) cells ✓

B. Smooth muscle

C. Nerve sheath

D. Fat

Correct: A. *They are adenocarcinomas of glandular origin.*

Q2. Which best describes the prostate's position?

A. Above the kidneys

B. Below the bladder, around the urethra ✓

C. Behind the stomach

D. Within the spleen

Correct: B. *This position explains urinary symptoms and examination access.*

SESSION 1

Knowledge check (2/3)

Q3. Stage and grade differ because:

- A. They mean the same thing
- B. Stage is extent and grade is aggressiveness ✓
- C. Grade is the extent
- D. Stage is the PSA value

Correct: B. *Stage is where and how far; grade is how aggressive the cells look.*

Q4. Prostate cancer most commonly spreads to:

- A. Bone ✓
- B. Skin
- C. Thyroid
- D. Tonsils

Correct: A. *Bone is the key site of distant spread.*

SESSION 1

Knowledge check (3/3)

Q5. A higher Grade Group generally means:

A. Less aggressive tissue

C. A lower PSA

B. More aggressive tissue ✓

D. An earlier stage

Correct: B. *Higher Grade Groups indicate more aggressive histology.*

SESSION 1

Session reflection

Enter your response in the text box. There is no single correct answer. These prompts are for your own practice and for discussion.

1 Why is it important for patients to understand that prostate cancer can present differently from person to person?

2 What is one important fact about prostate cancer in Rwanda that you would want a patient to understand?

3 How would you explain, in simple terms, what “stage” and “grade” refer to in prostate cancer?

Session 2

Screening and early detection

Follow Jean, 68, as you work through what screening is, who is at risk, and how to weigh the benefits and limitations of early detection.

SESSION 2

Learning objectives



Distinguish

Distinguish screening from evaluating a person with symptoms.



Identify

Identify established risk factors and concerning symptoms.



Weigh

Weigh the benefits and limitations of PSA-based early detection.



Discuss

Support a balanced, shared decision about testing.



Recognise

Recognise urgent red flags.

SESSION 2

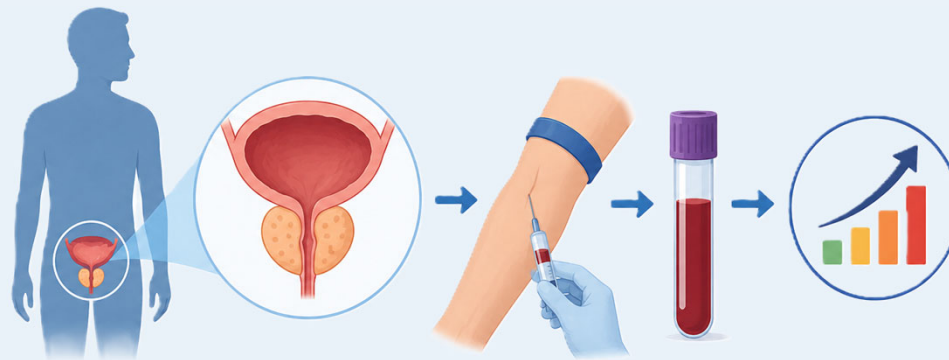
Case

SESSION 2

What is PSA (Prostate Specific Antigen)

PSA is a protein produced by the prostate and measured with a blood test. Levels can be affected by age, prostate infection or inflammation, prostate cancer, recent procedures, and some medications.

Recent ejaculation and vigorous exercise such as cycling may temporarily increase PSA and should ideally be avoided for 48 hours before testing. Finasteride and dutasteride can lower PSA and should be considered when interpreting results.



SESSION 2

What screening is

Screening

Early detection in a person with no symptoms. It often begins between age 40-50, depending on risk factor.



Symptomatic evaluation

Assessing a person who already has signs or symptoms. This is not screening.



Diagnostic testing

Testing after an abnormal result to work towards a diagnosis.



SESSION 2

Risk factors

Age

Risk rises with age.

Family history

A father or brother with prostate cancer raises risk.

Genetic risk

Some inherited factors raise risk where clinically relevant.

African ancestry

They develop prostate cancer at earlier ages and have more aggressive disease than other ethnicities



SESSION 2

Signs and symptoms

Often none

Early prostate cancer is often silent.

Urinary symptoms

Pain or burning while urinating, weak stream, hesitancy, frequency and nocturia are common but could be benign.

Blood in the urine or semen

Haematuria or haemospermia needs assessment.

Advanced disease

Bone pain can occur with more advanced disease.



⚠ Red flags

New leg weakness or numbness, difficulty walking, or loss of bladder or bowel control can signal spinal cord compression. Act urgently.

SESSION 2

Benefits and limitations of PSA

Possible benefit

Large randomized trials and long-term follow-up studies suggest that PSA-based screening can reduce prostate cancer-specific mortality.

Note: evidence comes predominantly from non-African populations. Large randomized controlled trials evaluating PSA screening in African populations are lacking.

Possible harms

- False-positive results
- Anxiety and worry
- Unnecessary further testing
- Overdiagnosis
- Overtreatment

Keep it balanced

PSA screening involves weighing potential benefits against potential harms. The decision should consider the individual patient's risk factors, preferences, and clinical context.

SESSION 2

Apply it to Jean

Scenario Jean has symptoms and is worried. He asks if a blood test will tell him whether he has cancer.

What is the most appropriate response?

- A** Tell him the PSA will confirm whether he has cancer
- B** Explain that a PSA can guide whether more assessment is needed, but cannot confirm cancer on its own, and discuss the benefits and limitations ✓
- C** Refuse to discuss testing
- D** Start treatment now

Correct answer: B

Why PSA is not cancer-specific. Offer a balanced discussion and shared decision and explain that an abnormal result leads to further assessment.

SESSION 2

Knowledge check (1/3)

Q1. Screening applies to:

- A. People with symptoms
- B. Asymptomatic people considering early detection ✓
- C. Only people already diagnosed
- D. No one

Correct: B. *Screening is early detection in people without symptoms.*

Q2. An elevated PSA:

- A. Confirms cancer
- B. May prompt further assessment ✓
- C. Means no action is needed
- D. Is always cancer

Correct: B. *PSA is not cancer-specific.*

SESSION 2

Knowledge check (2/3)

Q3. Which is an established risk factor?

A. Family history of prostate cancer ✓

C. Left-handedness

B. High water intake

D. Living at altitude

Correct: A. *Age, family history and African ancestry are established.*

Q4. A red flag needing urgent action is:

A. Mild nocturia

C. Occasional frequency

B. New leg weakness or loss of bladder or bowel control ✓

D. Mild hesitancy

Correct: B. *These can signal spinal cord compression.*

SESSION 2

Knowledge check (3/3)

Q5. A weak urinary stream:

- A. Confirms prostate cancer
- B. Has several possible causes and needs assessment ✓
- C. Is always cancer
- D. Needs no assessment

Correct: B. *Common urinary symptoms are often benign.*

SESSION 2

Session reflection

Enter your response in the text box. There is no single correct answer. These prompts are for your own practice and for discussion.

1 How would you open a balanced conversation about PSA testing with someone like Jean?

2 What would most influence whether you suggest further assessment?

3 Which misconception about PSA do you most often meet, and how would you address it?

Session 3

Referral and counselling

Turn assessment into the right next step: refer correctly, recognise emergencies, and counsel with care.

SESSION 3

Learning objectives



Refer

Refer according to urgency for concerning findings.



Recognise

Recognise urgent complications such as spinal cord compression.



Counsel

Counsel clearly about an abnormal result and the next step.



Support

Give basic psychological support.



Follow up

Encourage follow-up and complete the records.

SESSION 3

The referral pathway

Health post or centre

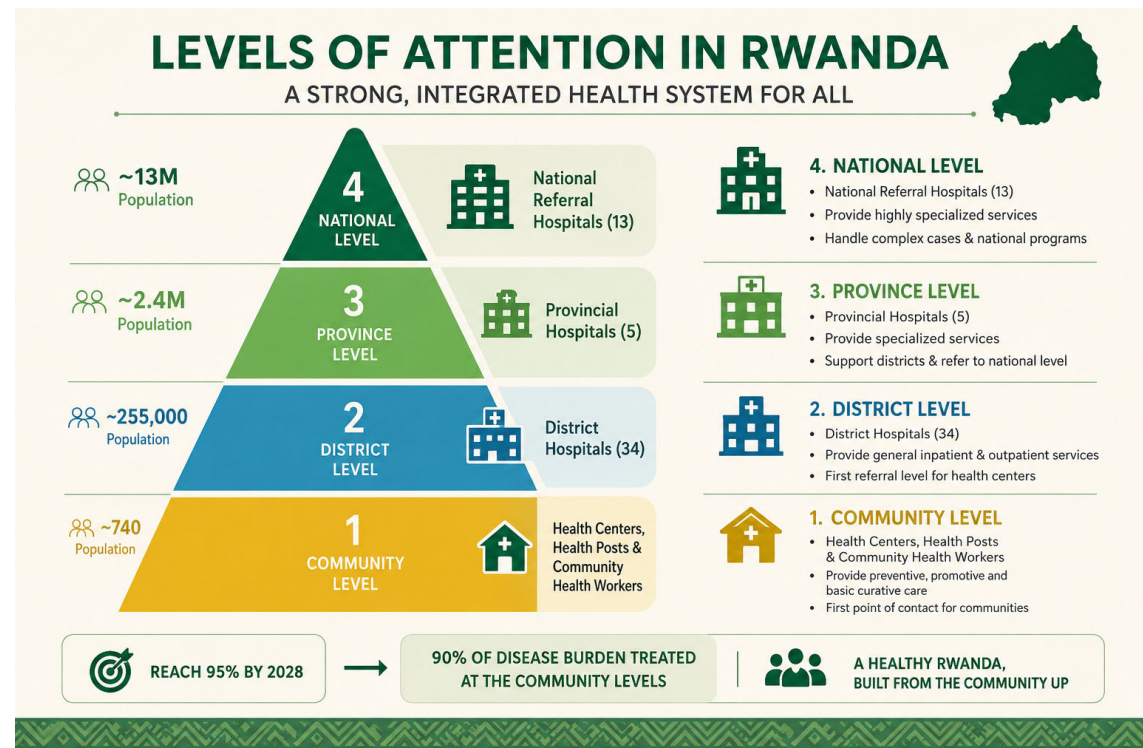
First contact: assess, record, and start the referral.

District hospital

Further assessment and onward referral where needed.

Referral centre

Specialist diagnosis, staging and treatment planning.



RBC CONFIRMATION REQUIRED Confirm the intended Rwanda referral route, the initial referral destination, and facility-level responsibilities for suspected prostate cancer. Also, check the number for each level as they might be different for 2026.

SESSION 3

How urgently to refer

Routine referral

For findings needing specialist assessment without time pressure.

Prompt referral

For concerning findings or an abnormal result that should be assessed soon.

Urgent or emergency

For suspected serious complications. Do not wait.
Arrange immediate assessment.

Possible cord compression (emergency)

New leg weakness or numbness
Difficulty walking
Loss of bladder or bowel control



RBC CONFIRMATION REQUIRED Confirm the Rwanda criteria that define routine versus prompt versus urgent referral.

SESSION 3

Counselling

Meaning of an abnormal result

Explain that it means more assessment is needed, not that cancer is certain.

Uncertainty

Acknowledge that the diagnosis is not yet known.

Prepare for referral

Explain what will happen next and why.

Fear and misconceptions

Address fear of cancer and correct myths without pressure.

Follow-up

Encourage attending the referral and returning for results.



SESSION 3

Psychological support

Listen

Listen without dismissing concerns.

Allow questions

Give time and answer plainly.

No false promises

Avoid premature reassurance.

When to escalate

Recognise when extra psychosocial support is needed.



SESSION 3

Jean's story continues

SESSION 3

Apply it

Scenario After you explain that his result needs further assessment, Jean says quietly that if it is cancer, nothing can be done, so why bother going.

What is the most helpful response?

- A** Agree that little can be done
- B** Acknowledge his fear, explain that assessment guides care and that options exist, and encourage him to attend ✓
- C** Tell him not to worry
- D** End the conversation

Correct answer: B

Why Acknowledge the emotion, correct the misconception gently, and support attendance without pressure.

SESSION 3

Knowledge check (1/3)

Q1. An abnormal PSA before diagnosis should be explained as:

- A. Confirmed cancer
- B. A signal that more assessment is needed ✓
- C. A reason to stop care
- D. A normal result

Correct: B. *It guides further assessment.*

Q2. Which is an emergency?

- A. Mild frequency
- B. New loss of bladder or bowel control with leg weakness ✓
- C. Occasional nocturia
- D. Mild hesitancy

Correct: B. *This can signal cord compression.*

SESSION 3

Knowledge check (2/3)

Q3. Good counselling before referral:

- A. Promises a diagnosis
- B. Explains the next step and allows questions ✓
- C. Withholds information
- D. Pressures the patient

Correct: B. *Explain respectfully and allow questions.*

Q4. The first point of contact in the pathway is usually:

- A. The cancer referral centre
- B. The health centre ✓
- C. A private pharmacy
- D. No facility

Correct: B. *First contact assesses, records and starts the referral.*

SESSION 3

Knowledge check (3/3)

Q5. Basic psychological support includes:

- A. Premature reassurance
- C. Ignoring fear

- B. Listening and allowing questions ✓**
- D. Rushing the patient

Correct: B. *Listen and give time.*

SESSION 3

Session reflection

Enter your response in the text box. There is no single correct answer. These prompts are for your own practice and for discussion.

1 How would you explain an abnormal PSA to a worried patient?

2 What might stop someone completing a referral, and how could you help?

3 How would you recognise and act on a possible spinal cord compression?

Session 4

Diagnosis and treatment

Understand what happens after referral: how a diagnosis is confirmed, and an overview of treatment and its effects.

SESSION 4

Learning objectives



Outline

Outline the diagnostic pathway to confirmation.



Explain

Explain the role and limits of examination and PSA.



Describe

Describe biopsy and histopathology for patient counselling.



Summarise

Summarise treatment categories and their purpose.



Correct

Correct common misconceptions.

SESSION 4

The diagnostic pathway

Clinical Assessment

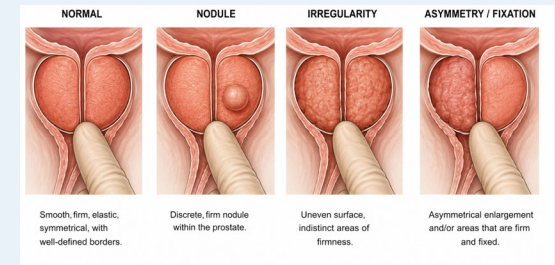
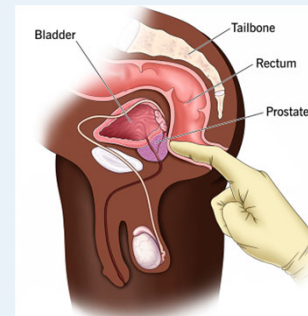
1. Medical history

- **Urinary symptoms:** frequency, urgency, nocturia, weak stream
- **Concerning symptoms:** hematuria, bone/back pain, weight loss
- **Risk factors:** age and family history of prostate/related cancers
- **Previous history:** PSA results, biopsies, prostate conditions
- **Overall health:** comorbidities, medications, functional status

2. Physical Exam

- **Digital rectal examination (DRE):** assess prostate size, symmetry, consistency and mobility.
- **Suspicious findings:** hard nodule, irregularity, asymmetry or fixation

A normal examination does not exclude cancer, and an abnormal one does not confirm it.



SESSION 4

The diagnostic pathway

Laboratory Tests

PSA

PSA LEVEL (ng/mL)		ACTION
	< 1.0	Low risk. Re-test in a few years.
	1.0 to 3.0	Average risk. Re-test in 1 to 2 years.
	Above the age limit up to ~10 (grey zone)	Repeat PSA, exclude reversible causes, then refer.
	> 10	Refer for specialist assessment.
	Any level with abnormal DRE or red flags	Refer regardless of the number.

AGE-SPECIFIC UPPER LIMITS (ng/mL)			
 40 y 2.5	 50 y 3.5	 60 y 4.5	 70 y 6.5

SESSION 4

Imaging

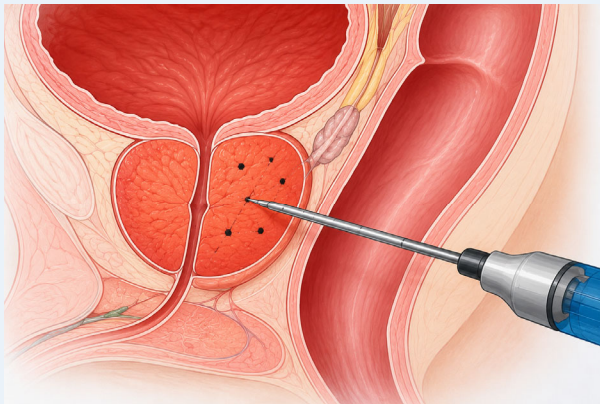
	ULTRASOUND (US) GUIDED BIOPSY	Guides a needle to collect prostate tissue samples (biopsy) for diagnosis.	HOW IT'S DONE: A probe in the rectum creates ultrasound images to guide a thin needle into the prostate to collect small tissue samples.
	MRI (MULTIPARAMETRIC MRI)	Finds suspicious areas, assesses tumor size, and helps guide biopsy.	HOW IT'S DONE: You lie in an MRI scanner. Magnets and radio waves create detailed images. No radiation. Takes ~30–45 minutes.
	PSMA PET/CT	Detects and evaluates spread of cancer to lymph nodes, bones, and other organs.	HOW IT'S DONE: A small amount of tracer is injected into a vein. After 1–2 hours, you lie on a table that slides into the scanner to detect “hot spots.”
	CT SCAN (COMPUTED TOMOGRAPHY)	Evaluates the entire body for metastatic disease, including lymph nodes, lungs, liver, adrenal glands, and other visceral organs.	HOW IT'S DONE: You lie on a table that moves through a CT scanner. X-rays create cross-sectional images of the entire body from neck to pelvis (or whole body). Intravenous contrast may be used to enhance detection of lesions.
	BONE SCAN	Detects areas of increased osteoblastic activity in the skeleton, typical of bone metastases.	HOW IT'S DONE: A small amount of tracer is injected into a vein. After 2–3 hours, a gamma camera scans your entire body to detect areas of increased activity in the bones.

Note: PET/CT services are not currently available but are planned for introduction in Rwanda.

SESSION 4

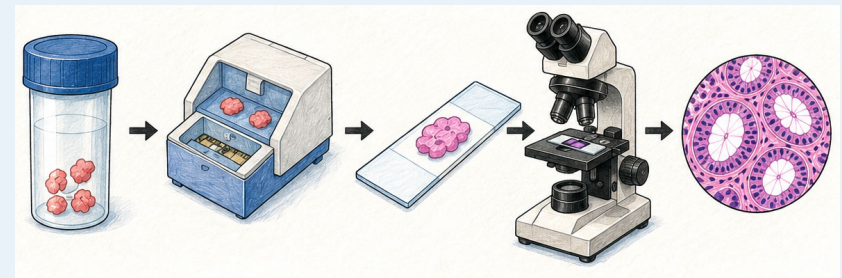
Biopsy

- Confirms the diagnosis of prostate cancer.
- Small tissue samples are collected from the prostate using a needle.
- Usually performed with ultrasound guidance.
- Explain that some discomfort, bleeding and infection risk can occur, and which warning signs need attention.



Histopathology

- Confirms prostate cancer by examining biopsy tissue under a microscope.
- Determines the type of prostate cancer.
- Assesses tumor differentiation and aggressiveness.
- Reports the Gleason score and Grade Group (1–5).
- Provides key information for risk assessment and treatment planning.



SESSION 4

Match the clue to the next step

In the module this is an interactive dropdown. The correct pairing is shown here.

Raised PSA in a well man considering early detection



Balanced discussion and shared decision about testing

Abnormal digital rectal examination



Prompt referral for specialist assessment

New leg weakness and loss of bladder control



Urgent referral for suspected spinal cord compression

Confirmed cancer, planning care



Imaging and biopsy results used for staging

Mild nocturia and PSA level of 3.1 in a 60-year-old healthy man



Counselling on signs and symptoms, recommend re-testing in 1-2 years

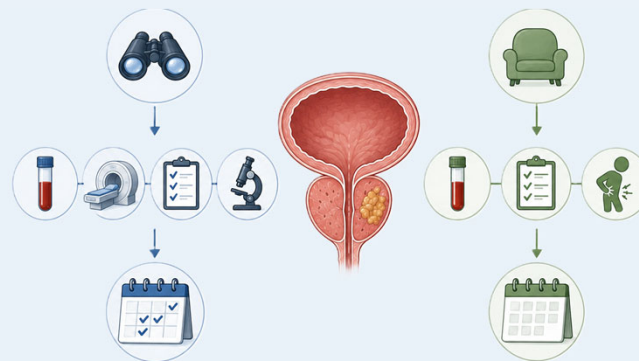
SESSION 4

Treatment overview

Note: Prostate cancer treatment decisions are made jointly by the patient and their specialist. The frontline HCP support education, referral, and continuity of care.

Non-active approaches

- **Active surveillance:** Regularly monitor localized, low-risk prostate cancer (e.g., Gleason score 6/Grade Group 1). These cancers may not pose a significant health concern, and immediate treatment may cause more harm than benefit. Treatment can be considered if the cancer shows signs of progression.
- **Watchful waiting:** Monitor the patient without regular testing aimed at detecting cancer progression. The focus is on managing symptoms if they develop rather than curing the cancer.

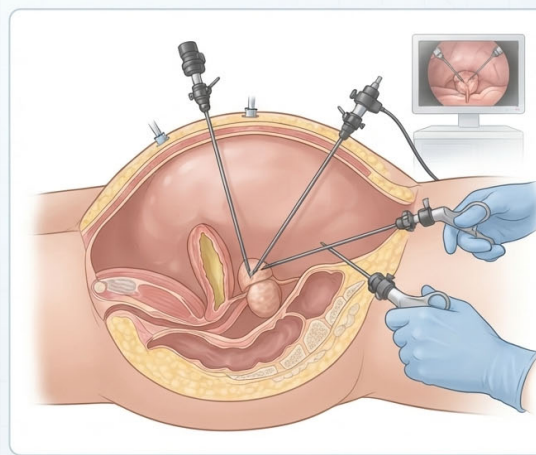
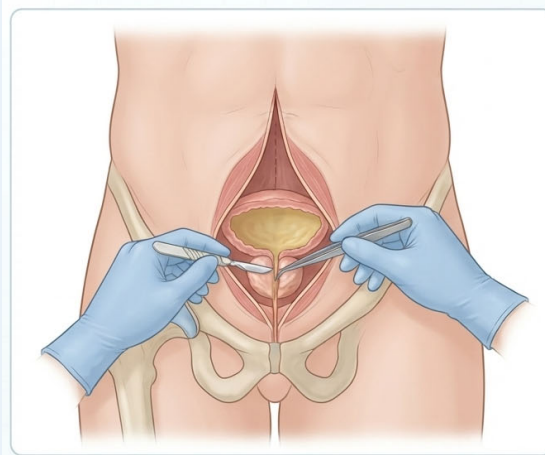


SESSION 4

Treatment overview

Surgery

- Radical prostatectomy removes the prostate and seminal vesicles to treat localized prostate cancer.
- **Techniques:** open, laparoscopic, or robot-assisted surgery.
- Lymph nodes may also be removed when indicated.
- Urinary urgency, frequency, and incontinence and erectile dysfunction are common after surgery



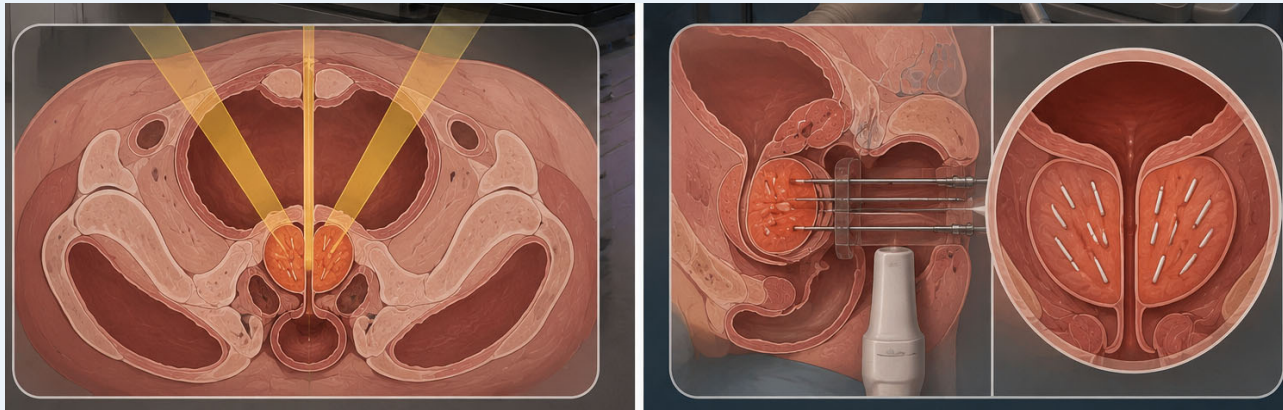
SESSION 4

Treatment overview

Radiotherapy

Uses radiation to destroy prostate cancer cells. May be combined with hormonal therapy.

- **External radiotherapy (EBRT):** radiation delivered from outside the body.
- **Brachytherapy:** radiation delivered from inside the prostate.
- Urinary frequency and urgency, bowel changes such as diarrhea or rectal irritation, fatigue, and erectile dysfunction are common after radiotherapy

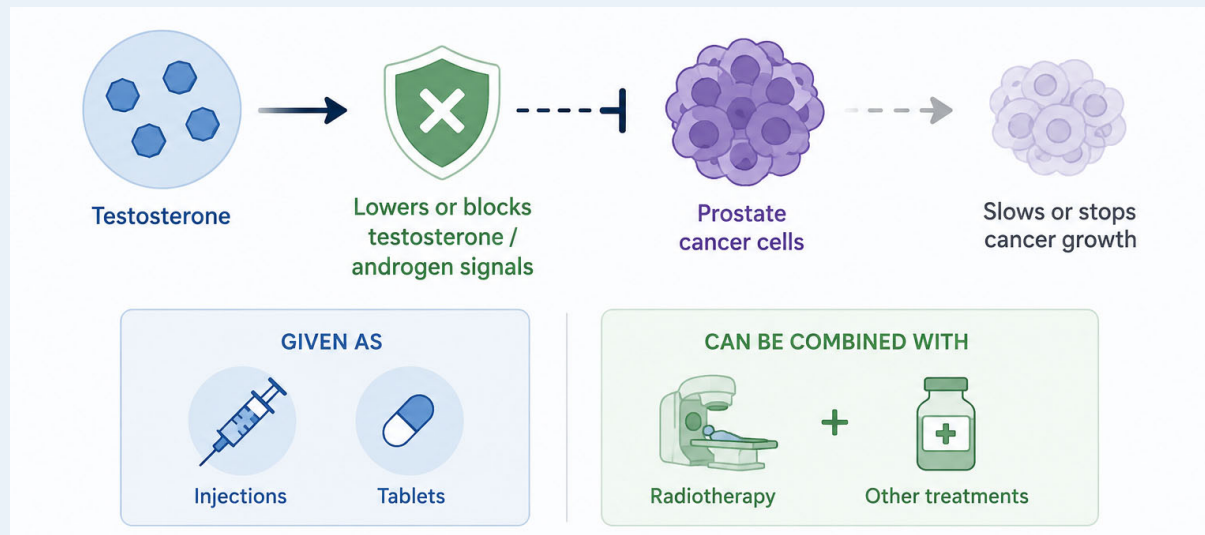


SESSION 4

Treatment overview

Hormonal therapy/ Androgen Deprivation Therapy

- Lowers testosterone or blocks androgen signaling, which prostate cancer cells use to grow. This includes traditional androgen deprivation therapy (ADT) and androgen pathway drugs. Mainly used for advanced or high-risk prostate cancer and may be combined with radiotherapy or other treatments.

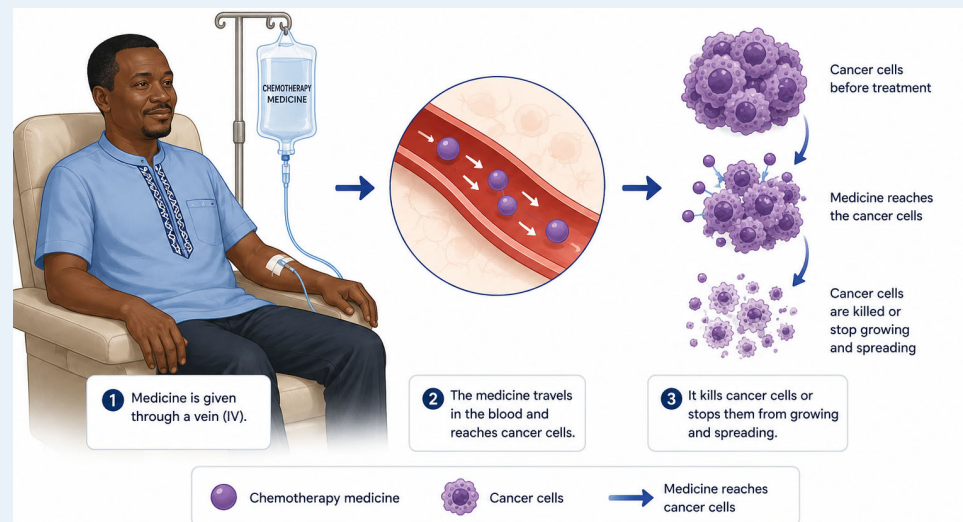


SESSION 4

Treatment overview

Chemotherapy

- Uses cytotoxic drugs to kill dividing cancer cells.
- Mainly for advanced or metastatic prostate cancer.
- Common drugs are taxanes (docetaxel, cabazitaxel).
- Given by drip in cycles, usually with hormonal therapy.

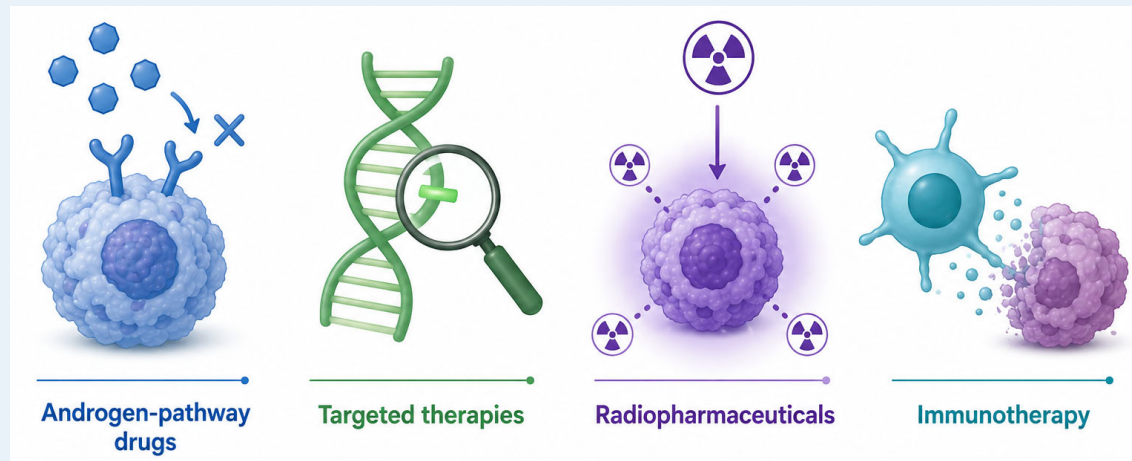


SESSION 4

Treatment overview

Targeted therapies

- Androgen-pathway drugs: Block androgen signaling more strongly to slow cancer growth.
- Targeted therapies: Target specific changes in cancer cells, such as DNA-repair defects.
- Radiopharmaceuticals: Deliver radiation directly to cancer cells or sites where the cancer has spread.
- Immunotherapy: Helps the immune system recognize and attack cancer cells.



SESSION 4

Common misconceptions

Urinary symptoms always mean cancer

False. They are common and often benign but deserve assessment.

A high PSA means cancer

False. PSA can rise for several reasons; it guides assessment.

No symptoms means no cancer

False. Early disease is often silent.

A diagnosis always means immediate treatment

False. Some cancers are monitored.

Biopsy spreads cancer

False. A prostate biopsy takes small tissue samples to confirm cancer and guide treatment. It can cause short-lived effects like a little blood or mild discomfort, but it does not make cancer spread. The real risk is avoiding it, which delays diagnosis.

SESSION 4

Apply it

Scenario A person has a raised PSA, an abnormal examination, and urinary symptoms. A family member asks whether the diagnosis is now certain.

What is accurate?

- ☐ A The diagnosis is confirmed
- ☒ B Cancer is not confirmed until biopsy and histopathology ✓
- ☐ C PSA alone confirms it
- ☐ D Symptoms confirm it

Correct answer: B

Why Symptoms, PSA and examination raise suspicion, but tissue confirms the diagnosis.

SESSION 4

Knowledge check (1/3)

Q1. Prostate cancer is confirmed by:

- A. PSA alone
- C. Biopsy and histopathology ✓**
- B. Symptoms alone
- D. Examination alone

Correct: C. *Tissue confirms the diagnosis.*

Q2. A normal digital rectal examination:

- A. Excludes cancer
- B. Does not exclude cancer ✓**
- C. Confirms cancer
- D. Replaces biopsy

Correct: B. *It does not exclude cancer.*

SESSION 4

Knowledge check (2/3)

Q3. Radiotherapy works by:

- A. Adding hormones
- B. Damaging cancer cell DNA with radiation ✓
- C. Being a type of surgery
- D. Doing nothing

Correct: B. *It damages DNA so cells cannot keep dividing.*

Q4. Androgen deprivation therapy:

- A. Adds testosterone
- B. Reduces or blocks androgen signalling ✓
- C. Is a type of surgery
- D. Only relieves pain

Correct: B. *Prostate cancer growth is often androgen-driven.*

SESSION 4

Knowledge check (3/3)

Q5. Active surveillance means:

- A. Ignoring the cancer
- B. Structured monitoring of suitable low-risk cancer ✓
- C. Immediate surgery
- D. Palliative care only

Correct: B. *It is structured monitoring, not neglect.*

SESSION 4

Session reflection

Enter your response in the text box. There is no single correct answer. These prompts are for your own practice and for discussion.

1 How would you explain a biopsy to a patient in plain terms?

2 How would you describe the difference between local and systemic treatment?

3 Which misconception would you most want to correct in your setting?

Session 5

Supportive treatment

Support people through treatment and beyond: recognise effects, know the emergencies, and protect quality of life.

SESSION 5

Learning objectives



Recognise

Recognise and sort common treatment effects.



Escalate

Escalate urgent warning signs.



Support

Support comfort and quality of life.



Follow up

Encourage follow-up and reporting of new symptoms.

SESSION 5

Adverse effects and care

Separate expected effects from those needing assessment, and from urgent warning signs.

Urinary effects

Leakage of urine, usually on effort, is common early and often improves over months with pelvic floor exercises.

Management: Pelvic floor exercises, continence advice; refer if persistent.



Sexual effects

Erectile dysfunction is common and often improves with support. Loss of ejaculate and natural fertility is permanent, so discussion on fertility preservation should occur beforehand.

Management:



SESSION 5

Adverse effects and care

Separate expected effects from those needing assessment, and from urgent warning signs.

Urinary effects

Leakage of urine

Management: Pelvic floor exercises, continence advice; refer if persistent.

Urinary urgency/frequency

Management: Check for UTI, bladder/fluid advice; refer if persistent.



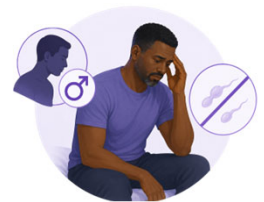
Sexual effects

Erectile dysfunction

Management: Provide education and support; GPs may prescribe PDE5 inhibitors when appropriate, with referral to specialist or psychological services if needed.

Loss of ejaculate and natural fertility

Management: Discuss fertility concerns and refer for specialist support; consider sperm banking before treatment.



SESSION 5

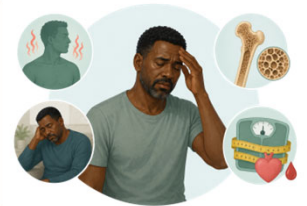
Adverse effects and care

Separate expected effects from those needing assessment, and from urgent warning signs.

Hormonal effects

Hot flushes, fatigue, and effects on bone and metabolism with androgen deprivation.

Management: Lifestyle and exercise support, monitoring of bone and metabolic health, and referral for troublesome symptoms.



Bowel and radiotherapy

Bowel effects and fatigue can follow radiotherapy.

Management: Provide hydration and dietary advice, encourage appropriate activity and rest, and refer persistent or significant bowel symptoms for specialist assessment.



Chemotherapy effects

Fatigue, nausea and vomiting, increased infection risk, and sometimes numbness or tingling in the hands and feet

Management: Support hydration, nutrition and fatigue management; reinforce prescribed anti-nausea treatment. Fever or signs of infection require urgent assessment and escalation.



SESSION 5

Jean has been treated for prostate cancer. He returns for a follow-up appointment.

These things have been really bothering me, but I feel awkward talking about them.

Jean shares his concerns.

I leak urine when I cough, sneeze or laugh. Sometimes I even leak before I get to the toilet. It's so frustrating and embarrassing.

And honestly... I'm also having trouble with erections. It's not like before and it's really upsetting.

A supportive response.

Thank you for telling me, Jean. You're not alone—many men experience changes like these after treatment.

Let's talk about how these are affecting you so we can find the right support.

Explaining that help is available.

There are options and support for both leaking urine and erectile changes.

Leaking urine
Pelvic floor exercises can help strengthen the muscles that control your bladder. We can also refer you to a specialist if needed.

Erection changes
There are treatments that can help, like tablets or other options. A specialist can talk with you about what might work for you.

Working together on a plan.

I'll refer you to our urology/oncology team and give you information and resources that can help.

We'll work together to help you feel better and get back to the things that matter to you.

SUPPORT IS AVAILABLE

Key takeaway.

- Ask in a respectful, open way.
- Listen without judgement.
- Explain that support and treatment options are available.
- Refer or coordinate with specialists when needed.

"Some men experience changes in bladder control or sexual function after treatment. Have you noticed any changes that you would like to talk about?"

SESSION 5

Palliative care

What it is

Relief of symptoms and support for quality of life, alongside treatment when appropriate.

Not only end of life

It is not limited to the last days of life.

In advanced disease

Helps with pain, bone pain, mobility, fatigue and distress.



SESSION 5

Quality of life

Function

Urinary, sexual and physical function matter.



Wellbeing

Emotional wellbeing, relationships and daily activities matter.



Respectful care

Discuss sexual and urinary health respectfully and without stigma.



Practical needs

Consider work, finances and follow-up.

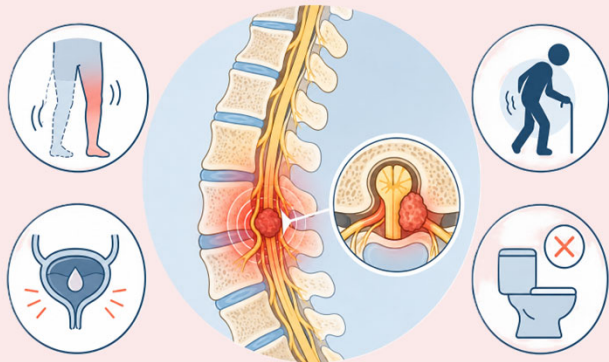


SESSION 5

Urgent warning signs

Possible cord compression

New or worsening leg weakness or numbness, difficulty walking, severe back pain, or loss of bladder or bowel control. This is an emergency and requires immediate assessment.



Severe symptoms

Severe or rapidly worsening pain, inability to pass urine, significant bleeding, fever or signs of serious infection, or sudden deterioration require urgent assessment.



SESSION 5

Apply it

Scenario A man on treatment reports new, severe back pain with weakness in both legs and difficulty passing urine.

What is the priority?

- A** Reassure and review next month
- B** Arrange urgent assessment for possible spinal cord compression ✓
- C** Advise rest only
- D** Increase fluids

Correct answer: B

Why These features suggest cord compression. Arrange urgent assessment and refer without delay.

SESSION 5

Knowledge check (1/3)

Q1. Palliative care is:

- A. Only for the last days of life
- B. Symptom relief and quality of life alongside treatment when appropriate ✓
- C. The same as stopping care
- D. Only pain medicine

Correct: B. *It focuses on symptoms and quality of life.*

Q2. An urgent warning sign is:

- A. Mild fatigue
- B. New leg weakness with loss of bladder control ✓
- C. An occasional hot flush
- D. Mild nausea

Correct: B. *This can signal cord compression.*

SESSION 5

Knowledge check (2/3)

Q3. Hot flushes and bone effects are linked to:

- A. Radiotherapy only
- B. Androgen deprivation therapy ✓
- C. Biopsy
- D. PSA testing

Correct: B. *These follow androgen deprivation.*

Q4. Sexual and urinary health should be discussed:

- A. Avoided as too sensitive
- B. Respectfully and without stigma ✓
- C. Only if the patient raises it
- D. Never

Correct: B. *Discuss them clinically and respectfully.*

SESSION 5

Knowledge check (3/3)

Q5. A new, worsening symptom should be:

- A. Ignored
- B. Reported and assessed ✓
- C. Delayed for months
- D. Self-treated only

Correct: B. *Encourage reporting and assessment.*

SESSION 5

Session reflection

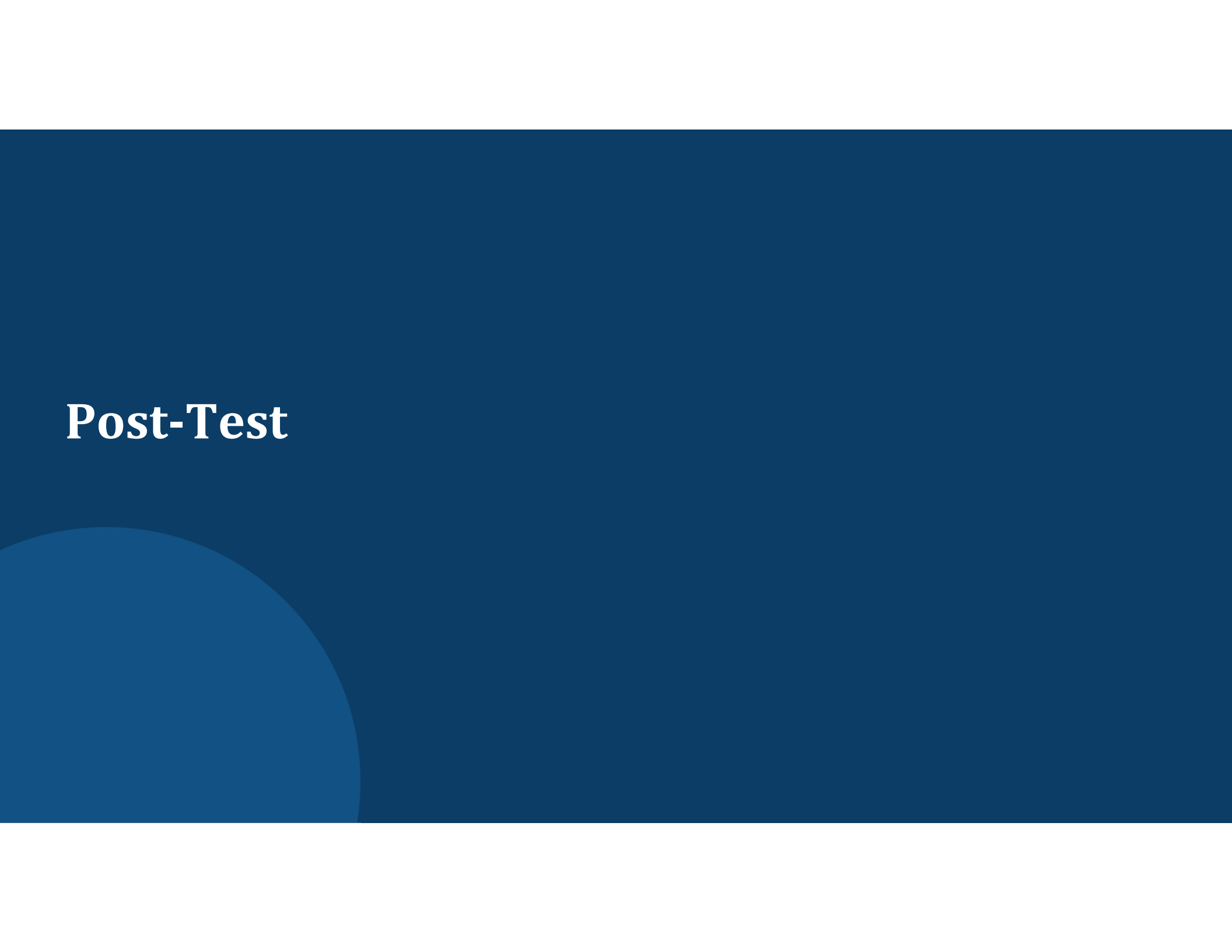
Enter your response in the text box. There is no single correct answer. These prompts are for your own practice and for discussion.

1 How would you help a patient sort which symptoms to report urgently?

2 How would you raise sexual or urinary health respectfully?

3 What support could help someone continue follow-up?

Post-Test

The image features a solid dark blue background. On the left side, there is a large, lighter blue circular shape that is partially cut off by the edge of the frame. The text "Post-Test" is written in a white, serif font, positioned in the upper left area of the image.

BASELINE CHECK

Knowledge check (1/10)

Q1. Most prostate cancers start in which cells?

- A. Muscle cells
- B. Glandular (epithelial) cells
- C. Nerve cells
- D. Blood vessels

Correct: B. *Most prostate cancers are adenocarcinomas, arising from glandular cells.*

Q2. Which is an established risk factor?

- A. Younger age
- B. Family history of prostate cancer
- C. High water intake
- D. Living at altitude

Correct: B. *Age, family history and African ancestry are established risk factors.*

BASELINE CHECK

Knowledge check (2/10)

Q3. Early prostate cancer usually causes:

- A. Severe pain
- B. Often no symptoms
- C. Immediate weight loss
- D. Fever

Correct: B. *Early disease is frequently silent.*

Q4. A raised PSA means:

- A. Cancer is confirmed
- B. Further assessment may be needed
- C. No further action
- D. A benign result only

Correct: B. *PSA is not cancer-specific; it guides further assessment.*

BASELINE CHECK

Knowledge check (3/10)

Q5. Urinary symptoms such as a weak stream:

- A. Confirm prostate cancer
- B. Can have benign causes
- C. Are always harmless
- D. Always mean infection

Correct: B. *Common urinary symptoms have several possible causes.*

Q6. Prostate cancer commonly spreads to:

- A. Bone
- B. Skin
- C. Hair
- D. Teeth

Correct: A. *Bone is a key site of distant spread.*

BASELINE CHECK

Knowledge check (4/10)

Q7. Stage describes:

- A. How aggressive the cells look
- B. Where the cancer is and how far it has spread
- C. The PSA value
- D. The treatment chosen

Correct: B. *Stage is extent; grade is microscopic aggressiveness.*

Q8. Grade (Gleason or Grade Group) describes:

- A. Extent of spread
- B. How abnormal the cells look
- C. The patient's age
- D. The PSA level

Correct: B. *Grade is microscopic aggressiveness.*

BASELINE CHECK

Knowledge check (5/10)

Q9. A normal digital rectal examination:

- A. Excludes cancer
- B. Does not exclude cancer
- C. Confirms cancer
- D. Replaces biopsy

Correct: B. *A normal examination does not exclude cancer.*

Q10. Prostate cancer is confirmed by:

- A. PSA alone
- B. Symptoms alone
- C. Biopsy and histopathology
- D. Examination alone

Correct: C. *Tissue confirms the diagnosis.*

BASELINE CHECK

Knowledge check (6/10)

Q11. Screening applies to:

- A. People with symptoms
- B. Asymptomatic people considering early detection
- C. Only people with cancer
- D. No one

Correct: B. *Screening is early detection in people without symptoms.*

Q12. Active surveillance means:

- A. Ignoring the cancer
- B. Structured monitoring of suitable low-risk cancer
- C. Immediate surgery
- D. Palliative care only

Correct: B. *It is structured monitoring, not neglect.*

BASELINE CHECK

Knowledge check (7/10)

Q13. A red flag needing urgent action is:

- A. Mild nocturia
- B. New leg weakness or loss of bladder or bowel control
- C. Occasional frequency
- D. Mild hesitancy

Correct: B. *These can signal spinal cord compression.*

Q14. Androgen deprivation therapy works by:

- A. Adding testosterone
- B. Reducing or blocking androgen signalling
- C. Killing all cells equally
- D. Only relieving pain

Correct: B. *Prostate cancer growth is often androgen-driven.*

BASELINE CHECK

Knowledge check (8/10)

Q15. Overdiagnosis means:

- A. Missing a cancer
- B. Detecting a cancer that would not have caused harm
- C. A wrong diagnosis
- D. A late diagnosis

Correct: B. *Some cancers grow so slowly they would not have caused harm in a lifetime.*

Q16. Radiotherapy:

- A. Uses ionizing radiation to damage cancer cell DNA
- B. Is only a type of surgery
- C. Adds hormones
- D. Has no side effects

Correct: A. *It damages DNA so cells cannot keep dividing.*

BASELINE CHECK

Knowledge check (9/10)

Q17. Which is often used to assess possible bone spread?

- A. Ultrasound of the neck
- B. Bone imaging
- C. Skin scan
- D. No imaging

Correct: B. *Bone is a common metastatic site.*

Q18. Counselling before referral should:

- A. Promise a diagnosis
- B. Explain that assessment is needed without creating fear
- C. Withhold information
- D. Avoid the patient's questions

Correct: B. *Explain the next step respectfully and allow questions.*

BASELINE CHECK

Knowledge check (10/10)

Q19. Palliative care is:

- A. Only for the last days of life
- B. Symptom relief and quality of life, alongside treatment when appropriate
- C. The same as stopping care
- D. Only pain medicine

Correct: B. *It focuses on symptoms and quality of life.*

Q20. In Rwanda, prostate cancer in men is:

- A. Rare
- B. The most common cancer in men in current estimates
- C. Only found abroad
- D. Not recorded

Correct: B. *In the GLOBOCAN 2024 estimates it is the leading cancer in men in Rwanda.*

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